MIL-T-40612B

6 February 1969
SUPERSEDING
MIL-T-40612A
22 April 1966

MILITARY SPECIFICATION

TYING MACHINES, BAG, INDUSTRIAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers automatic and manually operated bag tying machines that close and tie plastic, mesh, or paper bags.

* 1.2 Classification.-

1.2.1 Types.- Bag tying machines covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Automatic - Maximum of 60 bags tied per minute
Type II - Manual operated - Maximum of 30 bags tied per minute

1.2.2 Styles.- Type II bag tying machines shall be of the following styles:

Style 1 - For use with flattened wire
Style 2 - For use with wire reinforced plastic ribbon

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

CC-M-636 - Motor, Alternating-Current, (Fractional Horsepower).
PPP-B-601 - Boxes, Wood, Cleated Plywood.

FSC 3540

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

MILITARY

MIL-P-116 - Preservation; Methods of.
MIL-B-121 - Barrier-Material, Greaseproofed, Waterproofed, Flexible.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Electrical Construction Material List

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois, 60611; Walt Whitman Road, Melville, L. I., New York, 11746; or 1655 Scott Boulevard, Santa Clara, California, 95050).

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NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

Standard ICl-1959 - Industrial Controls

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, New York, 10017).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass., 02110).

(Technical society and technical association specifications and standards are generally available from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample tying machine shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all the applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standard product.- The bag tying machines shall be a current commercial product of the manufacturer except for any changes necessary to comply with this specification. All parts in any one contract having the same manufacturer's part number shall be functionally and dimensionally interchangeable for the same type and style tying machine.

3.3 Code and standards.-

3.3.1 Compliance.- The tying machine shall comply with the following:

(1) Electrical control components shall conform to NEMA Standard, ICl-1959 or to UL Standards, as applicable.

(2) The electrical materials and components shall conform to UL standards, as applicable; the installation of the electrical components shall conform to the applicable requirements of NFPA Standard, No. 70.

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3.3.2 Certification.- Prior to approval of the preproduction sample, or if none is submitted, prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the tying machine he proposes to furnish under this specification meet the requirements specified in 3.3.1.

3.3.2.1 NFPA and NEMA.- Acceptable evidence of meeting the requirements of NFPA Standard, No. 70 and NEMA Standard, No. ICl-1959 shall be the manufacturer's certified statement that the tying machine conforms to the applicable requirements.

3.3.2.2 UL.- Acceptable evidence of meeting the requirements of UL shall be the UL label, a listing in the UL "Electrical Construction Material List", or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the electrical materials and components offered have been tested and conform to the applicable UL standards.

* 3.4 Materials.- The materials shall conform to the applicable specifications and requirements specified herein. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

* 3.5 Design and construction.- Type I and II bag tying machines shall consist of an electric motor, a bag receiver and closer assembly that feeds, cuts off a measured band of flattened steel wire or reinforced plastic ribbon, as applicable, and wraps the band to close the bag.

* 3.5.1 Type I.- Type I bag tying machine shall have the motor, bag receiver and closer assembly as an integral bag closing head mounted on a vertical column with a stable base. The head shall be adjustable on the column and shall have means to lock the head at any height from not less than 12 to not more than 44 inches from the floor. This bag tying machine, requiring only that the top of the bag be held to remove the slack and to be guided into the receiver, shall automatically gather, fasten the neck with a wire band, and release the bag as the filled bag is being conveyed through the receiver. This machine shall close filled bags ranging in weight from 1 to 10 pounds, with plated, flattened steel wire 0.060 x 0.028 inches (see 6.5).

* 3.5.2 Type II, style 1.- Type II, style 1 bag tying machine shall be similar to the type I machine except that the bag receiver shall be closed at one end so that the filled bag shall be manually inserted at the opened end and withdrawn when the bag has been closed. This bag tying machine shall close filled bags ranging in weight from 1 to 10 pounds, with plated, flattened steel wire 0.060 x 0.028 inches (see 6.5).

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* 3.5.3 Type II, style 2.- Type II, style 2 bag tying machine shall be a bench mounted unit. This tying machine shall close bags by wrapping and twisting a piece of reinforced plastic ribbon around the neck as the bag is manually inserted and withdrawn from the bag receiver. The machine shall close the bag with a plastic ribbon 5/32 or 3/16 inches wide, reinforced with a strand of 0.018 inch diameter wire (see 6.6).

* 3.6 Electric motor.- Unless otherwise specified (see 6.2), the electric motor shall conform to CC-M-636 and shall operate from a 120-volt, 60 cycle, single phase, alternating current power source. A gear motor with motor conforming to CC-M-636 is acceptable.

* 3.7 Lubrication.- Type I and II bag tying machines shall have accessible means for the lubrication of moving parts requiring lubrication and seals to prevent the leakage of lubrication. At the time of delivery the machine shall be lubricated with the manufacturer's standard lubricant.

* 3.7.1 Lubricating chart.- A lubricating chart (metal plate or decal), indicating all points where lubrication and grade of lubrication is required and the time interval when lubrication shall be accomplished, shall be attached to each tying machine.

* 3.8 Performance.- Type I and II bag tying machines shall be easily loaded and threaded with the tying wire or plastic ribbon, as applicable, and shall operate smoothly in closing bags of produce and fruit at the filled weight and rate as indicated in table I, with no damage to the bag, when tested in 4.4.1.

TABLE I.- Performance data.

Bag contents	Weight	Rate Number of bags closed per minute		
		Type I	Type II, style 1	Type II, style 2
Carrots	3-lb. bag	60	30	20
Onions	3-lb. bag	60	30	20
Potatoes	10-lb. bag	40	20	15
Apples	5-lb. bag	60	30	15

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* 3.9 Electromagnetic compatibility.- When specified (see 6.2), type I and II tying machines shall be equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.2).

* 3.10 Equipment manual.- The supplier shall furnish with each bag tying machine an equipment manual. Manual contents and additional distribution shall be as specified in contract or order (see 6.2).

* 3.11 Marking for identification.- Bag tying machines shall be marked in accordance with the manufacturer's standard practice.

* 3.12 Special tools.- Special tools required for disassembly, maintenance, and adjustment of the machine shall be furnished with each bag tying machine.

3.13 Finish.- The bag tying machines shall be finished in accordance with the manufacturer's standard commercial practice. Painted surfaces shall be smooth, adherent, and free of blisters, runs, imbedded foreign matter, and areas of no coating.

* 3.14 Workmanship.- The complete bag tying machine shall be clean, new, properly assembled, and perceptibly in alignment. All exposed surfaces shall be free of burrs and sharp edges and show no evidence of rust or corrosion. Components shall not be loose, bowed, dented, or fractured.

* 3.14.1 Rivets.- Rivets shall not be fractured and shall be tight with heads completely formed and in full contact with the surface of the riveted member.

* 3.14.2 Fasteners.- Screw fasteners shall be drawn tightly; not fractured or stripped. Fasteners subject to excessive vibration shall be secured with antivibration means such as castellated nuts with cotter pin, lock washer and self-locking nuts.

* 3.14.3 Welding.- Welding shall be continuous, smooth, adherent, free from pinholes, fractures, slag inclusion, and flux deposits.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required it shall be examined for defects in table II, dimensions specified, and tested in accordance with 4.4.1 and 4.4.2, as applicable.

* 4.3 Inspection.- Sampling for inspection shall be made in accordance with the provisions set forth in MIL-STD-105, unless otherwise specified hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- A lot shall consist of all the tying machines of the same type and style offered for inspection at one time. The sample unit for this examination shall be one completely fabricated tying machine.

* 4.3.2.1 Visual examination.- Examination shall be made of the end item for the defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished in accordance with manufacturer's standard commercial practice.		X
	Painted surface not smooth or adherent, not free of blisters, runs, imbedded foreign matter, or areas of no coating.		X

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship, general (applicable to all components and assemblies)	Unit not clean.		X
	Not perceptibly in alignment.	X	
	Surfaces not free of burrs or sharp edges.		X
	Adjustable assembly that cannot be adjusted.	X	
	Components loose.		X
	Bowed, dented, or fractured.	X	
Rivets	Fractured.	X	
	Loose, head not fully formed or not in full contact with riveted surface.		X
Fasteners	Loose.		X
	Fractured or stripped.	X	
	Cotter pins of castellated nuts, lock washers, or lock nuts missing.		X
Welding	Not continuous, smooth, adherent, not free from pinholes or fractures.	X	
	Slag inclusion or flux deposits.		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminals or not inclosed in specified conduit where required.	X	
	Adequate slack not provided or required insulation stripped from wiring.	X	
Lubrication (where applicable)	Not lubricated where required.	X	
	Fitting not readily accessible for servicing.		X
Lubrication chart	Lubrication chart missing, incomplete, not legible.	X	

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Identification marking	Missing, incomplete, not legible, not as specified.		X
Equipment manual	Incomplete, not legible.		X
	Missing.		X

* 4.3.2.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions specified. Any dimension not within the specified requirements shall constitute a defect. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units.

* 4.3.2.3 Testing of the end item.- When a preproduction sample is not required, the initial production unit shall be tested in accordance with 4.4.1 and 4.4.2, as applicable. Any nonconformance shall be cause for rejection. In addition, the test specified in 4.4.1 shall be performed using an inspection level of S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

* 4.3.3 Code and standards compliance.- Proof of compliance with the requirements of 3.3.1 shall be made available to the Government representative.

* 4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table III. The lot shall be the number of containers offered for delivery at one time. The sample unit shall be one shipping container fully prepared for delivery. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examine	Defect
Marking (Exterior and interior)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.

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TABLE III.- Examination of preparation for delivery (cont'd)

Examine	Defect
Materials	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps; loose strapping or tape banding; inadequate stapling; bulging or distortion of containers; blocking, bracing, cushioning, case liners inadequate, improper, or missing.
Contents	Tying machine equipment parts, manual, or special tools (if applicable) missing.

4.4 Tests.-

* 4.4.1 Performance test.- The bag tying machine shall be loaded with the specified steel wire or reinforced plastic ribbon, as applicable, and operated for 5 minutes tying bags of each variety of produce and fruit specified in table I to determine compliance with 3.8 (see 6.3).

* 4.4.2 Electromagnetic compatibility test.- When electromagnetic compatibility is required, the preproduction sample or initial production unit shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his testing organization. The supplier shall furnish the contracting officer, for approval, the Interference Control Plan, the EMI/EMC (Electromagnetic Interference/ Electromagnetic Compatibility) Test Plan, and the Electromagnetic Compatability Test Report required in MIL-STD-461 (see 6.4). Upon approval of the test report by the contracting officer and provided that all other requirements of the specification are met, the preproduction samples or initial unit of production, as applicable, shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.-

5.1.1.1 Disassembly.- The closing head and base of the machine shall be disassembled from the vertical column when possible and when such will reduce the cube of the pack.

5.1.1.2 Preservation.- Preservation shall be in accordance with Method I of MIL-P-116. All exposed, uncoated ferrous metal surfaces of each machine and any special tools shall be cleaned process C-1, thoroughly dried, and coated with preservative conforming to type P-9 of MIL-P-116.

5.1.1.3 Protection of preservative.- Surfaces from which preservative compound might be removed by contact with packaging and packing materials shall be covered or wrapped with greaseproof barrier-material conforming to type I or II, grade A of MIL-B-121. Barrier-material shall be secured with tape, conforming to class 2 or 3 of PPP-T-45, cord, or twine or heat sealed when heat sealable barrier material is used.

5.1.1.4 Packaging.- The electric cord shall be compactly coiled and secured to the machine with cord, twine, or tape.

* 5.1.1.5 Equipment manual.- The equipment manual shall be prepared method IC-1 of MIL-P-116, and secured to the machine.

* 5.1.2 Level C.- Machines shall be preserved and packaged to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use or controlled humidity storage. The supplier may use his standard practice when it meets these requirements.

5.2 Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- One machine, complete with special tools, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to class 2, style 2 or 4, type 3 load of PPP-B-621; or overseas type, style A, B, or I, type 3 load of PPP-B-601. Lumber blocking and bracking shall be used to immobilize the contents. The blocking and bracing shall not depend on end-grain nailing alone. Supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. Each shipping container shall be provided with a type I or II, grade C, case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

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* 5.2.2 Level B.- One machine, complete with special tools, preserved and packaged as specified in 5.1, shall be packed in a shipping container conforming to class 1, style 2 or 4, type 3 load of PPP-B-621; or domestic type, style A or B of PPP-B-601. Lumber blocking and bracing shall be used to immobilize the contents. The blocking and bracing shall not depend on end-grain nailing alone. Supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing.

* 5.2.3 Level C.- Machines complete with special tools, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking.- Shipping containers shall be marked with arrows and the word "UP" in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The bag tying machines are intended for use in commissaries to close and tie bags of fruit and produce.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style required as applicable (see 1.2).
(c) First article - When preproduction sample is not required (see 3.1).
(d) Electrical requirements, if other than specified (see 3.6).
(e) Whether electromagnetic compatibility is required (see 3.9).
(f) Manual contents and distribution (see 3.10).
(g) Applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2).

* 6.3 Waiver of performance test.- In lieu of the test specified in 4.4.1 and at the option of the contracting officer, the supplier may submit a certified test report stating that a model identical to that

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offered has been tested or observed in normal operation in a commercial establishment for a minimum period of 5 minutes and conforms to the requirements specified herein. The model shown shall have been in operation for a period of not less than 30 days at the commercial establishment.

* 6.4 Electromagnetic compatibility.- The contracting officer should submit a copy of the Interference Control Plan, The EMI/EMC Test Plan, and Test Report required (see 4.4.2) to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey, 07703, for approval.

* 6.5 The plated steel wire "Silverstitch" by ACME Steel Company, Chicago, Illinois, has been found to be satisfactory for type I and II, style 1 bag tying machines.

* 6.6 The reinforced plastic ribbon by Plas-Ties Corporation, Santa Ana, California, has been found to be satisfactory for type II, style 2 bag tying machine.

6.7 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Review activities:

Army - SM, CE, MS
Navy - SA, MC
Air Force - 84

Preparing activity:

Army - GL

Project No. 3540-0059

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Tying Machines, Bag, Industrial MIL-T-40612B

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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